

Represent Chance with Fractions, Decimals, or Percentages

Step 7

 Upper Elementary
 (Grades 3–4)

 Today I am learning to represent chance
 with fractions, decimals, or percentages.

1. Write as a Fraction

Write the chance as a fraction of the total.

a) Chance of red


 Fraction: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

b) Chance of green


 Fraction: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

c) Chance of blue


 Fraction: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

d) Chance of yellow


 Fraction: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

2. Write as a Decimal

Write the chance as a decimal.

a) Chance of red


 Decimal: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

b) Chance of yellow


 Decimal: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

c) Chance of green


 Decimal: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

d) Chance of blue


 Decimal: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$

3. Write as a Percentage

Write the chance as a percentage.

a) Chance of blue


 Percentage: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$

b) Chance of red


 Percentage: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$

c) Chance of green


 Percentage: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$

d) Chance of yellow


 Percentage: $\frac{\quad}{\quad} = \frac{\quad}{\quad}\%$

4. Convert Between Fraction, Decimal and Percentage

Complete the table.

	Fraction	Decimal	Percentage	Picture (out of 10)
a)	$\frac{3}{10}$	_____	30%	
b)	_____	0.7	_____%	
c)	$\frac{1}{2}$	_____	_____%	
d)	_____	0.25	_____%	
e)	$\frac{4}{5}$	_____	_____%	
f)	_____	0.9	_____%	

5. Real-Life Chance

Look at each situation. Write the chance as a fraction, decimal and percentage.

a) It might rain tomorrow.



Fraction	Decimal	Percentage
$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}\%$

b) Chance of drawing a heart from a standard deck of cards.



Fraction	Decimal	Percentage
$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}\%$

c) Chance of getting tails when flipping a coin.



Fraction	Decimal	Percentage
$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}$	$\frac{\quad}{\quad}\%$

7. Make, Record and Represent

Spin the spinner 20 times. Record your results and represent each chance.



Colour	Tally	Number	Fraction (out of 20)	Decimal	Percentage
Total		20			

6. Compare the Chances

Use <, > or = to compare the chances.

a)	$\frac{1}{4}$	$\bigcirc$	0.2	c)	60%	$\bigcirc$	$\frac{3}{5}$
b)	0.75	$\bigcirc$	$\frac{2}{3}$	d)	0.33	$\bigcirc$	$\frac{1}{3}$
e)	25%	$\bigcirc$	0.30	f)	$\frac{7}{10}$	$\bigcirc$	0.65

8. Think and Explain

- a) Which colour happened more often than you expected? Why?
- b) Which colour happened less often than you expected? Why?
- c) How can knowing fractions and decimals help us make better predictions?



I can...

- ☐ represent chance using fractions.
- ☐ represent chance using decimals.
- ☐ represent chance using percentages.
- ☐ compare and explain chances.
- ☐ use representations to make predictions.

Chance Words



Great job!

 You are using fractions,
 decimals and percentages
 to understand chance!